Theory Questions Report

Name: Natalie Sameh Saad

Code: 2500617

Report: Questions answers

Date: 19/08/2026

1- Perception Sensors:

1- Radars:

I. Working principle and Data Capture: The radar operates by transmitting radio waves towards the environment. When these waves hit obstacles, they bounce back to the sensor. By measuring the time-of-flight and Doppler frequency shift from the reflected signals, the system calculates the object's distance, speed and angle.

II. Type of Data: It captures numerical data representing the distance (range), relative speed, and angle/direction of objects in the environment.

III. Data processing Requirements: The raw signals undergo Signal Filtering to remove any environmental noise or unwanted reflections. Then, Kalman Filters are used to track moving objects and predict their future positions and velocities accurately over time.

2- LiDAR:

I. Working principle and Data Capture: LiDAR emits laser light into the environment. By using the known speed of light and measuring the time-of-flight (the time it takes for the reflected light to return), the system precisely calculates distances.

II. Type of Data: It captures three-dimensional representations known as Point Clouds (X, Y, Z coordinates) defining the geometry of objects in the surrounding environment.

III. Data Processing Requirements: The processing pipeline involves multi-scan data frames, Ground Removal to filter out ground points, and Clustering to accurately isolate detected obstacles.

3- Cameras:

I. Working principle and Data Capture: Cameras act as optical sensors where lenses focus the visible light reflected from the environment onto an image sensor to generate continuous visual frames.

II. **Type of data:** It captures rich RGB image frames containing detailed visual information, colors, textures, and semantic features (such as lane markings, traffic signs, and cones).

III. **Data Processing Requirements:** The captured frames first undergo image pre-processing and lens correction followed by models like YOLO to detect objects and outline them with bounding boxes, then the system detects specific keypoints on the objects, and finally, uses 3D pose estimation techniques to calculate the precise 3D position and orientation of the objects in the environment.

2- Stereo Vision:

1- 2 cameras capture the same scene at the same time from slightly different horizontal positions , after detection of object the location in the left image is slightly shifted compared to the right and that is called disparity.
2- Calculating of exact depth of the object using the measured disparity and physical distance between the two camera lenses and focal length using triangulation geometry.
3- Processing all the pixels then the system will convert 2D images to 3D position Data for the objects.
4- To make the corresponding pixels from the both cameras on the same horizontal axis rectification is applied.
5- We use SIFT to extract unique features from each view and KNN pairs them together in both images.

3- Machine Learning:

1- Definition of neural networks: They are machine learning models inspired by the biological neural networks that make up human brain as it’s considered a type of artificial intelligence .The artificial neurons organized into layers: an Input Layer that receives raw data, multiple Hidden Layers that perform the main work and an Output Layer that produces the final prediction or classification.

2- How do they learn: They take data, make a prediction, measure the error using a Loss Function, and then use Optimization (like Gradient Descent) to adjust their weights and reduce errors over time.

3- Processing sensor data for environment perception:

- Camera Processing: Using YOLOv2 and CNNs to analyze images, extract features, and draw bounding boxes around objects.
- LiDAR Processing: NN process LiDAR point clouds to cluster scattered points & reconstruct 3D shapes of objects.
- Sensor Fusion: Combining camera and LiDAR data together to give the system a complete and accurate understanding and prediction of objects in the environment.

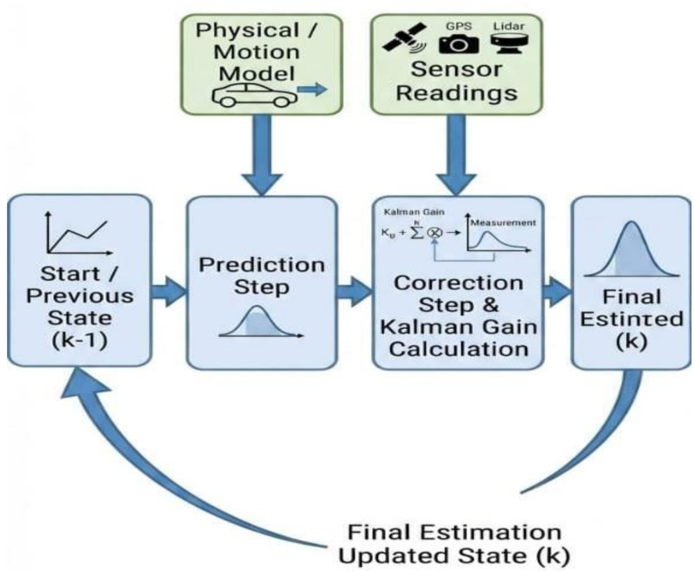
4- Kalman Filter:

1- Definition: It is an algorithm used to track moving objects by combining noisy sensor measurements with mathematical predictions to accurately estimate their true position and velocity over time.

2- How it can be used in vehicle localization:

- Prediction Step: The system uses a motion model to predict the vehicle's next position and state based on previous velocity and controls.
- Correction / Update Step: The system takes actual measurements from sensors (like GPS or wheel encoders) to correct the prediction error and find the true state.

3- EKF: Standard Kalman filters only work for linear systems, but real-world vehicle movements and sensor equations are non-linear. Therefore, the EKF (Extended Kalman Filter) is used because it linearizes non-linear models making it ideal for tracking and localization in autonomous vehicles and SLAM.



4- Localizing a self-driving car using a kalman filter:

- States to estimate as (position(x,y),velocities(v_x,v_y), acceleration(a_x,a_y), heading angle, yaw rate).
- Sensors to use as (IMU ,GNSS, GSS, Lidar and cameras).

5- SLAM:

1- The Definition: It stands for Simultaneous Localization and Mapping, which is process used by an autonomous vehicle to build a map of an unknown environment while simultaneously tracking its own precise location within that map.

2- Types of SLAM:

- Graph-based SLAM (as LSD-SLAM)
- EKF-SLAM
- Particle filter-based SLAM (FastSLAM 2.0).

3- Building map in SLAM: Sensors detect cone positions (landmarks) relative to the car. Observations are fused with velocity estimates using an Extended Kalman Filter as the motion model.

SLAM performs data association, updates landmark positions probabilistically, and flags unreliable landmarks so it builds a map of this track. Used sensors are LiDAR, Cameras, GNSS, IMU, GSS .

4- Localization of vehicle: In an unknown environment, the vehicle lacks a pre-existing map. Localization is performed simultaneously with mapping by continuously estimating the vehicle's motion and updating its position relative to observed landmarks (cones).

- In EKF-SLAM: Localization comes from continuously updating this state using sensor measurements (cone observations + velocity), but computational cost grows quadratically with the number of landmarks too slow for a track full of many cones in real-time.
- In Fast SLAM 2.0: Uses particles to track trajectory hypotheses and mini-maps, updating via prediction, weighting, and resampling. It scales, look-alike cones effortlessly (ideal for racing).

6-Graph Search Algorithms:

1- BFS: It explores the graph level by level using a Queue (FIFO). It visits all immediate neighbors before moving to the next level, while tracking visited nodes to avoid cycles.
Pros: It is both complete and optimal, meaning it is guaranteed to find a path to the goal if one exists and will always find the shortest path in unweighted graphs.
Cons: It Consumes a lot of memory because the queue stores all nodes of the current level at the same time.

2- DFS: Explores as deep as possible along each branch before backtracking using a Stack (LIFO) or recursion, while tracking visited nodes to prevent infinite loops.

Pros: It uses low memory because the stack only stores the current path.

Cons: It doesn't guarantee finding the shortest path, and can get trapped in infinite branches.

3- Dijkstra's: It finds the shortest path in a weighted graph by looking at actual values (like time or distance) instead of just counting steps blindly. Using a Priority Queue, it picks the node with the lowest current cost, explores its neighbors, and updates their distances step by step.

Pros: It guarantees finding the actual shortest path because it accounts for real edge weights.

Cons: It lacks a directional sense towards the goal, causing it to explore uniformly in all directions, which can take extra time.

4- A* algorithm: Combines Dijkstra's actual cost with a heuristic estimate to the goal .Using a Priority Queue, it evaluates nodes with the lowest total cost(heuristic estimate + actual cost) guiding the search directly toward the target instead of exploring blindly in all directions.

Pros: It's fast & Optimal as Highly efficient and guarantees finding the shortest path (provided the heuristic is admissible).

Cons: It's Heuristic Dependent as its performance and accuracy heavily depend on the quality of the chosen heuristic function and consumes large memory.

7-Practical Motion Planning:

1- The reason of occurence of Zig-Zag: The A* algorithm operates on a grid/cell structure, restricting the robot's movement to rigid directions (up, down, left, right, and diagonals), which forces the robot to move in rough paths (Zig-Zag) that waste battery and energy.

2- The solutions:

- Path Smoothing (Post-processing): Convert the sharp, angular corners into smooth, continuous curves using mathematical tools like Bezier Curves or Splines.
- Theta* Algorithm: Use any-angle path-planning algorithm instead of standard A*. It uses a "Line of Sight" check to allow direct, smooth diagonal movements without being restricted to grid lines.

3- Risks of applying these solutions:

- Collision Risk: Smoothing the path or cutting corners can cause the robot to drift too close to walls or obstacles, risking a crash.
- Increased Computation Time: Algorithms like Theta* or post-processing curve-fitting require higher processing power and calculation time.

8-Control Theory:

1- Control: The process of managing, directing, or regulating the behavior of a system to achieve a desired output.

2- Control Theory: The science of making systems or machines behave the way we want, keeping them stable and accurate in a safe way avoiding external problems. It’s important in engineering as it designs automated systems to work accurately and stay stable and never lose control when external problem happens.

3- Its applications:

Car Cruise control: It keeps the car automatically at a steady speed.
Air Conditioners: it monitors room temperature to keep it at the chosen setpoint.

4- Open-Loop Control System: A system where the control action is independent of the output (no feedback).
Example: A toaster heats bread for a set time and doesn’t see if the bread is burnt or undercooked.

5- Closed-Loop Control System: A system that uses feedback to compare the actual output with the desired goal and automatically makes corrections by calculating errors.

6- Feedback: is the mechanism of taking the system's output back to the controller to compare it with the required goal to reduce error and maintain accuracy.

7- Application on Feedback: A light sensor continuously monitors ambient lighting and automatically turns the headlights on when driving into dark areas or at sunset, acting as a smart control system that adjusts to the environment without requiring manual driver intervention.

9-PID Control:

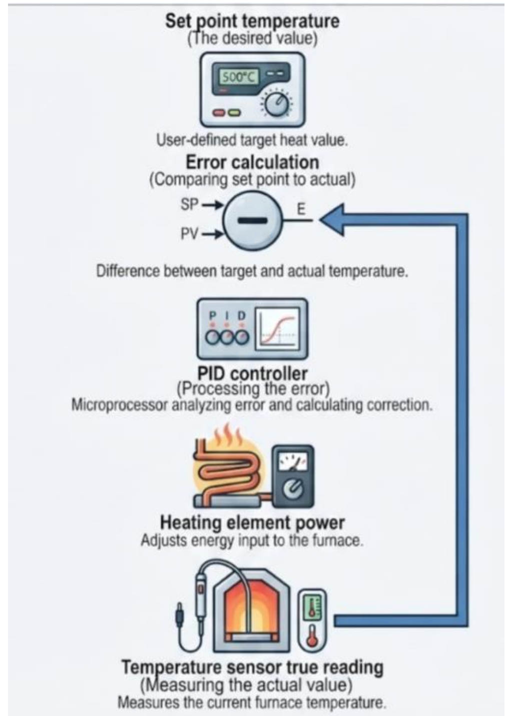
1- Definition: A feedback control mechanism used to maintain system's output at a desired setpoint as it calculates the error between them continuously in a closed loop

- Proportional(P):it deals with the current error, as the output is directly proportional to the error.
- Integral(I): It accumulates past errors to get rid of steady-state error formed by proportional term.
- Derivative(D): it deals with future error rate of change as it predicts future error to reduce overshoot.

2- Tuning Methods:

- Manual tuning: Set I,D to zero ,increase P to start oscillation then add D to reduce overshoot then add I to eliminate steady-state error to reach the target.
- Zeigler-Nichols Method: setI,D to zero , increase P until constant oscillations asre formed. Get the ultimate gain(K_u) and measure periodic time of oscillation (T_u),then use Zeigler-Nichols formulas to get Ideal values of P,I,D.

3- PID controller for an industrial furnace: I will determine the setpoint temperature , the temperature sensor will measure the actual value and sending it to PID controller to be compared to desired value so it will calculate the error between them , finally heating power element will reach desired value.



4- Limitations:

(D) terminal is sensitive to the noise and PID in general struggles with nonlinear systems as it is a linear control system.It is limited to single-input and single-output .

5- Better controller: MPC uses maths models to predict future system behavior with managing multiple variables in complex and non-linear processes.

10-References:

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7- https://youtu.be/zaBhtODEL0w?si=DpFq9juY_0cuSL8N

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